

Lab Report

Experiment NO: 01

Title of the experiment: Introduction to MATLAB:

Student ID: 23-54242-3

Student Name: Md. Zahid Hasan

Course Title: Data Communication

Section: L

Date of submission:

Objective: The main objective of this experiment is to gain knowledge about the use of MATLAB for solving various types of communication engineering problems and to develop understanding of MATLAB environments, commands and syntax.

Lab Data sheet

Exp-1: Introduction to MATLAB

Student Name - Md. Zahid Hasan

ID - 23-54242-3

Task-1: Entering matrices using built in-function
 $A [1\ 2\ 3; 8\ 6\ 4; 3\ 6\ 9]$

Solⁿ:

code: $\gg A = \begin{bmatrix} 1 & 2 & 3; \\ 8 & 6 & 4; \\ 3 & 6 & 9 \end{bmatrix}$

output: $A = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 6 & 4 \\ 3 & 6 & 9 \end{bmatrix}$

Task-2: Addressing an element

Solⁿ:

code: $\gg A(1,3) + A(2,1) + A(3,2)$

output: ans = 17

Task-3: K-th to J-th element

Solⁿ:

code: $\gg A(2:3, 1:2)$

output: ans = $\begin{bmatrix} 8 & 6 \\ 3 & 6 \end{bmatrix}$

Zahid
13/02/23

Task-4: Addressing the first two elements first row

Sol: code: `>> A(1, 1:2);`

output: ans = 1 2

Task-5: Address all the elements of second ~~row~~ column

Sol: code: `>> A(:, 2)`

output: ans =
2
6
6

Task-6: Generating matrices

Sol: code: `>> x = linspace(0, pi, 101)`

output:

FILE	VARIABLE				CODE	SARISK	ENVIRONMENT			RESOURCES				
C:\WINDOWS\system32														
Columns 1 through 14														
0	0.0314	0.0628	0.0942	0.1257	0.1571	0.1885	0.2199	0.2513	0.2827	0.3142	0.3456	0.3770	0.4084	
Columns 15 through 28														
0.4398	0.4712	0.5027	0.5341	0.5655	0.5969	0.6283	0.6597	0.6912	0.7226	0.7540	0.7854	0.8168	0.8482	
Columns 29 through 42														
0.8796	0.9111	0.9425	0.9739	1.0053	1.0367	1.0681	1.0996	1.1310	1.1624	1.1938	1.2252	1.2566	1.2881	
Columns 43 through 56														
1.3195	1.3509	1.3823	1.4137	1.4451	1.4765	1.5080	1.5394	1.5708	1.6022	1.6336	1.6650	1.6965	1.7279	
Columns 57 through 70														
1.7593	1.7907	1.8221	1.8535	1.8850	1.9164	1.9478	1.9792	2.0106	2.0420	2.0735	2.1049	2.1363	2.1677	
Columns 71 through 84														
2.1991	2.2305	2.2619	2.2934	2.3248	2.3562	2.3876	2.4190	2.4504	2.4819	2.5133	2.5447	2.5761	2.6075	
Columns 85 through 98														
2.6389	2.6704	2.7018	2.7332	2.7646	2.7960	2.8274	2.8588	2.8902	2.9217	2.9531	2.9845	3.0159	3.0473	
Columns 99 through 101														
3.0788	3.1102	3.1416												

Task: 7 =^A

Sol: Code: `>> v = (10:-2:0)`

output: ans = 10 8 6 4 2 0

Task-8: Increment by 1

Solⁿ: Code: `>> w = (5:10)`

output: `w = 5 6 7 8 9 10`

Task-9: Using Matlab function to provide zero

Solⁿ: code - `>> B = zeros(3,4)`

output: `B =`

0	0	0	0
0	0	0	0
0	0	0	0

Task-10: Using matrices ones formula

Solⁿ: code; `>> c = ones(2,5)*6`

output: `c =`

6	6	6	6	6
6	6	6	6	6

Task-11: Generates uniformly distributed random elements.

Solⁿ: code, `>> D = rand(1,5)`

0.8147	0.9058	0.1270	0.9134	
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output: `0.8147 0.9058 0.1270 0.9134 0.6324`

Task-12: Generating Gaussian-distributed random elements

Solⁿ: code;

output:

-1.3077	3.5784	3.0349
-0.4336	2.7694	0.7254
0.3426	-1.3499	-0.0631

Task-13: Deleting Rows and Columns

Solⁿ: code: $\gg A(2,:) = []$

output: $A =$

1 2 3
3 6 9

Task-14: Transpose of an array

Solⁿ: code: $\gg a = 0:3$
 $\gg b = a'$

output: $a = 0 \ 1 \ 2 \ 3$

$b =$
0
1
2
3

Task-15: Scalar application on matrix

Solⁿ: code: $\gg m = [1 \ 2 \ 3 \ 4; \ 5 \ 6 \ 7 \ 8; \ 9 \ 10 \ 11 \ 12]$
 $\gg 2 * m - 1$

output: $c =$

1	2	3	4
5	6	7	8
9	10	11	12

$ans =$

1	3	5	7
9	11	13	15
17	19	21	23

Task-16: Array+Array mathematics

Solⁿ: code

① >> d = [1 2 3; 4 5 6];

② >> e = [2 2 2; 3 3 3];

③ >> f = d + e

④ >> g = 2*d - e

⑤ >> h = d.*e

⑥ >> d./e

⑦ >> e.\d

⑧ >> c = A*B

⑨ >> A = [1 2 3; 4 5 6]

⑩ >> B = [1 2; 3 4; 5 6]

⑪ >> c = A*B

output: ① d =

1	2	3
4	5	6

② e =

2	2	2
3	3	3

③ f =

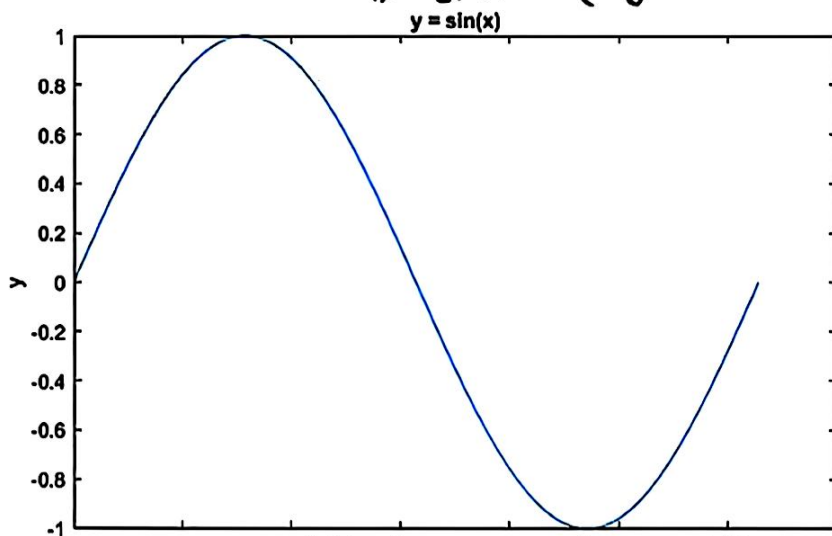
3	4	5
7	8	9

$$\textcircled{4} \begin{matrix} g = 0 & 2 & 4 \\ & 5 & 7 & 9 \end{matrix}$$

$$\textcircled{5} \begin{matrix} h = 2 & 4 & 6 \\ & 12 & 15 & 18 \end{matrix}$$

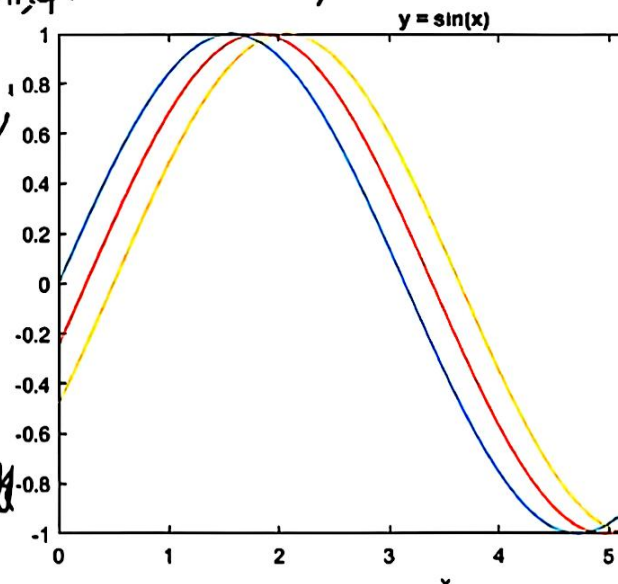
Task-17: Add title to the graph

Solⁿ: code:
 $\gg \text{xlabel}('x');$
 $\gg \text{ylabel}('y');$
 $\gg \text{title}('y = \sin(x)')$



Task-18: multiple graphs with a single call to plot

Solⁿ: code;
 $\gg x_1 = 0 : \pi/100 : 2 * \pi;$
 $\gg y_1 = \sin(x_1);$
 $\gg y_2 = \sin(x_1 - 0.25);$
 $\gg y_3 = \sin(x_1 - 0.5);$
 $\gg \text{plot}(x_1, y_1, x_1, y_2, x_1, y_3);$



Task-20: Using Controlling Axis

Soln: code:

```
>> t = -pi : pi/100 : pi;
```

```
>> s = cos(t);
```

```
>> plot(t,s)
```

```
>> axis([-pi pi -1 1])
```

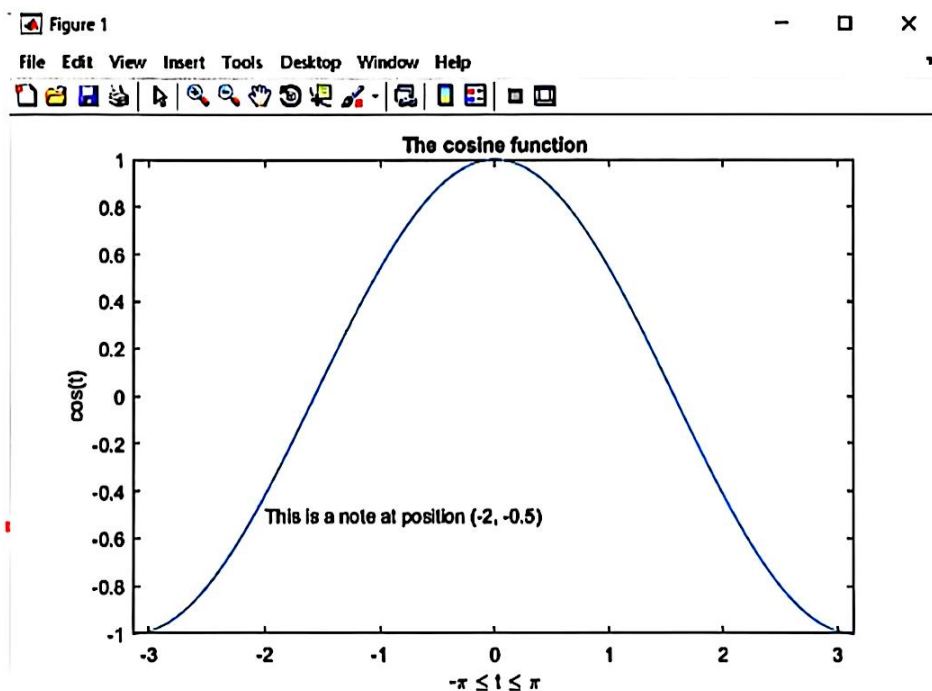
```
>> xlabel('-\pi \leq t \leq \pi')
```

```
>> ylabel('cos(t)')
```

```
>> title('The cosine function')
```

```
>> text(-2, -0.5, 'this is a note at position (-2, -0.5)')
```

/leq is used to generate the less-equal sign.



To

LAB Control Room

Faculty of Engineering

American International University-Bangladesh

Subject: Application for Makeup of First Lab Experiment

Forwarded for consideration
(Dr. Md. Humayun Kabir)
18/02/2026

Respected Sir,

With due respect, I would like to state that I have enrolled in the Data Communication course during the add-drop period of this semester. Due to my late enrollment, I was unable to attend the first lab experiment.

I am now attending all regular classes and laboratory sessions. Therefore, I kindly request you to grant me permission to complete the makeup of the first lab experiment at a scheduled time as per your convenience.

I shall remain grateful for your kind approval.

Yours sincerely,

Md. Zahid Hasan

Student ID: 23-54242-3

Department: Computer Science and Engineering

Semester: 8th

Section: L

Date: 18-02-2026

Datacom - L, Exp: 01

Room: DS308

Time: 8:30am.

Date: 19/02/2026

Thursday

Ref
01/02


19/02/26